4 Introduction to the Theory of Elasticity

The theory of elasticity is the study of the mechanics of continuous media, or in simple words, the deformation of the elements of a solid body by applied forces. In this chapter, we deal with static (time-independent) elasticity involving homogeneous deformations. In fact, the parameters defined here can also be used at the finite frequencies occurring in ultrasonic propagation. This is the simplest case and enables us to define concepts such as deformation, strain tensor, stress tensor, and the moduli of elasticity. We introduce tensor notation to describe the elastic parameters; it is a simple, elegant, and powerful approach that is used throughout advanced treaties in elasticity and acoustics. Complete discussions are given for tensors by Nye [1] and for elasticity by Landau and Lifshitz [2].

4.1 A SHORT INTRODUCTION TO TENSORS

Study of physics and engineering leads to categorizing measurable quantities as scalars or vectors. Scalars are physical quantities that can be represented by a simple number, for example, temperature. Equally important, they are not associated with direction. A vector, on the other hand, explicitly depends on direction, for example, velocity $\vec{V}$. In 3D space, we must specify the three components V_x, V_y, and V_z to describe the velocity vector $\vec{V}$ fully.

The concept of tensor has been introduced as an extension of the idea of a vector. In anisotropic media, tensors are essential to describe the relation between two vectors. But even in isotropic media, the idea of physical quantities specified by more than three components is essential, as will be seen in the theory of elasticity.

The concept of a tensor can be made concrete by a simple example—the electrical conductivity in a solid. For a 1D system (wire), it is customary to represent the conductivity σ as the proportionality constant linking the current density J to the electric field E, $J = \sigma E$. However, for a 3D medium that is anisotropic, the electric field $\vec{E}$ and the current density $\vec{J}$ will be, in general, in quite different directions. So, in general, one must write

$$\begin{aligned} J_1 &= \sigma_{11}E_1 + \sigma_{12}E_2 + \sigma_{13}E_3 \\ J_2 &= \sigma_{21}E_1 + \sigma_{22}E_2 + \sigma_{23}E_3 \\ J_3 &= \sigma_{31}E_1 + \sigma_{32}E_2 + \sigma_{33}E_3 \end{aligned} \tag{4.1}$$

Thus, to specify the conductivity fully, we need to specify the nine components that are usually written in matrix form as follows:

$$\sigma_{ij} = \begin{matrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{matrix} \tag{4.2}$$

The notation on the left is the tensor notation; for obvious reasons, σ_{ij} is termed a tensor of the second rank.

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For an isotropic system, $\vec{J}$ is always parallel to $\vec{E}$ and $|\vec{J}| = \sigma|\vec{E}|$. It follows in this case that the conductivity tensor is given by

$$\sigma_{ij} = \begin{matrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & \sigma \end{matrix} \tag{4.3}$$

A simple rule that follows from the general form of σ_{ij} is that the rank of a tensor is given by the number of indices. Thus, a scalar is a tensor of rank zero and a vector is a tensor of rank one.

At this point, it should be emphasized that although a tensor can be written in matrix form, it is not just a simple matrix. A tensor represents a real physical quantity, such as conductivity, whereas many matrices (e.g., change of coordinates) are simple mathematical relationships. Many advanced texts show that a tensor is rigorously defined by the way that it transforms under coordinate transformation (e.g., see Nye [1]), which will not be needed here, as all the tensors used in this book represent well-known physical properties.

From a practical standpoint, much economy of presentation and elegance can be obtained by using the Einstein convention. This convention says quite simply that when a suffix occurs twice in the same term, this automatically implies summation over that suffix, which becomes a dummy index or dummy suffix.

For example, Equation 4.1 can be written as

$$\begin{aligned} J_1 &= \sum \sigma_{1j} E_j \\ J_2 &= \sum \sigma_{2j} E_j \\ J_3 &= \sum \sigma_{3j} E_j \end{aligned} \tag{4.4}$$

or again

$$J_i = \sum_{j=1}^{3} \sigma_{ij} E_j$$

With the Einstein convention

$$J_i = \sigma_{ij} E_j \tag{4.5}$$

where it is understood, and never indicated explicitly, that i, j go over all available values, here 1, 2, and 3 or x, y, and z. In this relation, i gives the direction of current flow.

4.2 STRAIN TENSOR

The basic idea is that forces will be applied to solid bodies to deform them. As a starting point, there is a need to describe the deformation. If a point at $\vec{r}$ from the origin is displaced to position $\vec{r}$ by the force, then the deformation $\vec{u} = \vec{r}_i - \vec{r}$ is called the displacement vector. In tensor notation, $u_i = x_i - x$ where u_i and x_i are functions of u_i.

Since a point is displaced during a deformation, then the distance dl between two points close together is also changed. Using

$$\mathrm{d}l^2 = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 = \mathrm{d}x_i^2 \quad \text{before deformation} \tag{4.6}$$

$$\mathrm{d}l^{\,2} = \mathrm{d}x_i^{\,2} \quad \text{after deformation} \tag{4.7}$$

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Hence

$$\mathrm{d}l^{\,2} = (\mathrm{d}x_i + \mathrm{d}u_i)^2 \tag{4.8}$$

Using

$$\mathrm{d}u_i = \frac{\partial u_i}{\partial x_k}\,\mathrm{d}x_k \tag{4.9}$$

$$\mathrm{d}l^{\,2} = \mathrm{d}l^2 + 2\frac{\partial u_i}{\partial x_k}\mathrm{d}x_i\,\mathrm{d}x_k + \frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_l}\mathrm{d}x_k\,\mathrm{d}x_l \tag{4.10}$$

This can be written as follows:

$$\mathrm{d}l^{\,2} = \mathrm{d}l^2 + 2S_{ik}\,\mathrm{d}x_i\,\mathrm{d}x_k \tag{4.11}$$

where

$$S_{ik} = \frac{1}{2}\;\frac{\partial u_i}{\partial x_k} + \frac{\partial u_k}{\partial x_i} + \frac{\partial u_l}{\partial x_i}\frac{\partial u_l}{\partial x_k} \tag{4.12}$$

If the strains are sufficiently small, which will always be assumed to be the case in linear ultrasonics, then the quadratic terms can be ignored. The strain tensor S_{ik} is then

$$S_{ik} = \frac{1}{2}\;\frac{\partial u_i}{\partial x_k} + \frac{\partial u_k}{\partial x_i} \tag{4.13}$$

By construction, the strain tensor is symmetric so that the nine terms reduce to six. Clearly, three of these are diagonal and three are nondiagonal. Each diagonal term ($i = k = 1$, 2, or 3) has the simple significance shown in Figure 4.1. For example,

$$S_{11} = \frac{\partial u_1}{\partial x_1} \tag{4.14}$$

is clearly the extension per unit length in the x_1 direction. Hence, the diagonal terms correspond to compression or expansion along one of the three axes.

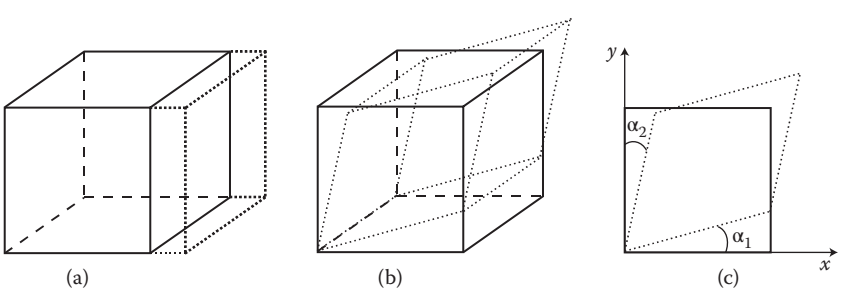

FIGURE 4.1 Strains for a unit cube. (a) Tensile strain u_{xx}. (b) Shear strain u_{xy}. (c) Definition of angles for shear strain u_{xy}.

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The off-diagonal terms can be understood with reference to Figure 4.1 for the case of a deformation of the plane perpendicular to the z axis. For small deformations

$$\tan\alpha_1 \sim \alpha_1 = \frac{\partial u_y}{\partial x}, \qquad \tan\alpha_2 \sim \alpha_2 = \frac{\partial u_x}{\partial y} \tag{4.15}$$

where α_1 and α_2 are angles with the x and y axes, respectively.

Thus, the change in angle between the two sides of a rectangle

$$\alpha_1 + \alpha_2 = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \tag{4.16}$$

is proportional to the shear strain S_{xy}.

A final property of the strain tensor can be obtained from the following mathematical results:

1. Any symmetric tensor can be diagonalized at a point by the choice of appropriate axes. If this is done, then the strain tensor has diagonal components $S^{(1)}$, $S^{(2)}$, and $S^{(3)}$ and the off-diagonal terms are zero.
2. The trace (i.e., the sum of the diagonal terms) of a symmetric tensor is invariant under change of coordinates. From (1), the trace will then be $S^{(1)} + S^{(2)} + S^{(3)}$ for the choice of any coordinate system.

Suppose that the coordinates are chosen so that S_{ik} is diagonal, then

$$\begin{aligned} \mathrm{d}l^2 &= (\partial_{ik} + 2S_{ik})\,\mathrm{d}x_i\,\mathrm{d}x_k \\ &= (1+2S^{(1)})\,\mathrm{d}x_1^2 + (1+2S^{(2)})\,\mathrm{d}x_2^2 + (1+2S^{(3)})\,\mathrm{d}x_3^2 \end{aligned} \tag{4.17}$$

where the relative displacement along axis i is $S^{(i)}$.

Consider the volume before and after deformation of a small volume element dV. It follows that

$$\begin{aligned} \mathrm{d}V &= \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3 \\ \mathrm{d}V &= \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3 \end{aligned} \tag{4.18}$$

so that

$$\begin{aligned} \mathrm{d}V &= \mathrm{d}V(1+S^{(1)})(1+S^{(2)})(1+S^{(3)}) \\ &\approx \mathrm{d}V(1+S^{(1)}+S^{(2)}+S^{(3)}) \end{aligned} \tag{4.19}$$

again neglecting the quadratic terms.

In any coordinate system, the trace can be written as follows:

$$S_{ii} = S_{11} + S_{22} + S_{33} \tag{4.20}$$

This gives finally

$$\mathrm{d}V = \mathrm{d}V(1+S_{ii}) \tag{4.21}$$

so that S_{ii} gives the relative change in volume under deformation. This can be expressed as the dilatation S, which is the change in volume per unit volume, which can be expressed as

$$S = S_{ii} = S_{11} + S_{22} + S_{33} \tag{4.22}$$

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4.3 STRESS TENSOR

We assume a body in static equilibrium under external forces such that there is no net translation or rotation. What is of concern is the effect of internal forces on a hypothetical unit cube inside the solid. These forces could arise, for example, from an ultrasonic wave impinging on the region in question. In principle, there could be two types of forces acting on the cube: (1) body forces (acting on the volume), and (2) surface forces. Body forces such as gravity will not be considered, so that a description is needed for surface forces acting on the faces of the cube. These forces will lead to deformation of the cube, which can be described by the strain tensor treated previously. Once this description has been obtained, we can formulate a 3D equivalent of Hooke's law for a relation between the forces and the deformations.

As shown in Figure 4.2, an applied force will generally be at some arbitrary angle to the unit cube. Since we are considering forces on the faces of the cube, we consider a particular face, for example, the *xy* face with normal along the *z* axis.

The components of the applied force can be classified into two major classes:

1. Normal component to the face, which will give rise to compressive or tensile stresses.
2. Tangential components, giving rise to shear stresses. For the example considered here are two of these: dF_x and dF_y.

In one dimension, the stress on a rod is defined as the force per unit area. In extending this definition to three dimensions, as above, clearly, there are two vectors involved: (1) the direction of the surface normal, and (2) the direction of the force. It follows that in three dimensions, the stress must be described by a stress tensor of rank two.

Extending directly from one dimension

$$T_{zz} = \frac{F_z}{A_z}, \quad T_{zy} = \frac{F_z}{A_y}, \quad T_{zx} = \frac{F_z}{A_x} \tag{4.23}$$

so that all of the components are described by a stress tensor of rank two.

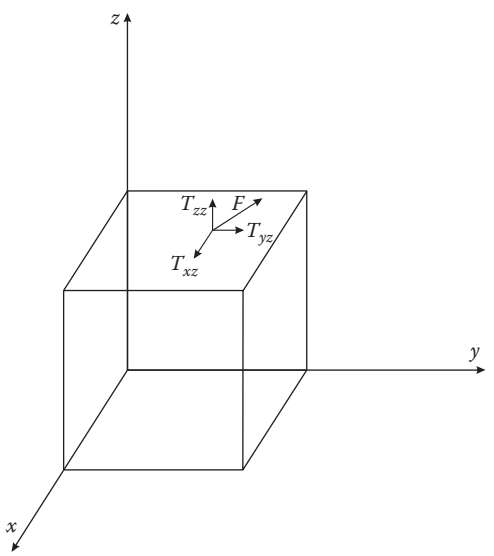

FIGURE 4.2 Definition of components of the stress tensor.

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The condition of static equilibrium leads to symmetry of the stress tensor. The tensile stresses along any one axis must balance; otherwise, the body would accelerate, so that there can only be three independent diagonal stresses. Likewise, the shear stresses must balance to avoid rotation, leading to three off-diagonal stresses. An elegant demonstration of this, together with a more abstract presentation of the stress tensor, is given by Landau and Lifshitz [2].

Finally, for the specific case of a liquid, the pressure is hydrostatic; it is uniform and same in all directions. Hence, for a sphere, the force in direction i on surface element dA is

$$\begin{aligned} \mathrm{d}F_i &= -p\mathrm{d}A_i = -p\delta_{ik}\,\mathrm{d}A_k \\ &= T_{ik}\,\mathrm{d}A_k \end{aligned} \tag{4.24}$$

Here, δ_{ij} is the Kronecker delta, an extremely useful mathematical device. It is defined as

$$\delta_{ik} = \begin{matrix} 1 & i = k \\ 0 & i \neq k \end{matrix} \tag{4.25}$$

Hence, for uniform hydrostatic compression

$$\begin{aligned} T_{ik} &= -p\delta_{ik} \\ T_{ik} &= \begin{matrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{matrix} \end{aligned} \tag{4.26}$$

The nondiagonal elements correspond to shear stress; these are zero, being in agreement with the fact that an inviscid liquid cannot support a shear stress.

4.4 THERMODYNAMICS OF DEFORMATION

Assume small and slow deformations so that the latter can be assumed to be elastic (so that it returns to its original state when external forces are removed) and reversible in the thermodynamic sense.

In general, the thermodynamic identity gives

$$\mathrm{d}U = T\mathrm{d}S + \mathrm{d}W \tag{4.27}$$

where

U = internal energy
T = temperature
S = entropy
W = work done on the system

For the particular case of hydrostatic compression

$$\mathrm{d}W = -p\,\mathrm{d}V = -p\,\mathrm{d}S_{ii} = -p\delta_{ik}\,\mathrm{d}S_{ik} = T_{ik}\,\mathrm{d}S_{ik} \tag{4.28}$$

so that

$$\mathrm{d}U = T\mathrm{d}S + T_{ik}\mathrm{d}S_{ik} \tag{4.29}$$

as shown elsewhere [2], this form is in fact true in the general case.

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For the Helmholtz free energy, $F = U - TS$, so

$$\mathrm{d}F = -S\mathrm{d}T + T_{ik}\mathrm{d}S_{ik} \tag{4.30}$$

and for the Gibbs free energy, $G = H - TS$

$$\begin{aligned} H &= U + pV \\ G &= U - TS - T_{ik}\mathrm{d}S_{ik} = F - T_{ik}\mathrm{d}S_{ik} \end{aligned} \tag{4.31}$$

From the form of a perfect differential in terms of its partial derivatives

$$T_{ik} = \left.\frac{\partial U}{\partial S_{ik}}\right|_S = \left.\frac{\partial F}{\partial S_{ik}}\right|_T \tag{4.32}$$

and

$$S_{ik} = -\left.\frac{\partial G}{\partial T_{ik}}\right|_T \tag{4.33}$$

4.5 HOOKE'S LAW

In its simplest form, Hooke's law states that for small elongations of an elastic system, the stress is proportional to the strain. There are two different and equivalent approaches to Hooke's law for an isotropic solid, each important and instructive in its own way. The first [2] is based on Landau's classic expansion of the Helmholtz free energy in parameters of the system and subsequent application of statistical physics. In this case, the free energy F is expanded in terms of the strain tensor:

$$F_T = F_0 + \frac{1}{2}\lambda u_{ii}^2 + \mu S_{ik}^2 \tag{4.34}$$

where λ and μ are called the Lamé coefficients. This expansion takes into account the following points:

1. For the undeformed system at constant temperature, $S_{ik} = 0$ and $T_{ik} = 0$. Since $T_{ik} = \left(\partial F/\partial S_{ik}\right)_T$, there is no linear term in the expansion.
2. Since F is a scalar, every term in the expansion must be a scalar. Since the diagonal terms S_{ii}^2 and all diagonal terms S_{ik}^2 are scalars, the coefficients λ and μ are also scalars.

The form of F can be rewritten to take into account the two fundamentally different forms of deformation of isotropic bodies:

1. Pure shear, corresponding to constant volume and change in shape. $S_{ii} = 0$ in this case.
2. Pure hydrostatic compression, corresponding to change in volume at constant shape.

Any deformation can be written as the sum of these two forms, leading to the following form:

$$S_{ik} = \left(S_{ik} - \frac{1}{3}\delta_{ik}S_{ll}\right) + \frac{1}{3}\delta_{ik}S_{ll} \tag{4.35}$$

The first term is pure shear and the second hydrostatic compression.

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The free energy can be rewritten to show shear and compression explicitly as follows:

$$F = \quad S_{ik} - \frac{1}{3}\delta_{ik}S_{ll} \quad ^{2} + \frac{1}{2}KS_{ll}^{2} \tag{4.36}$$

where now

$$\begin{aligned} K &= \lambda + \frac{2}{3} \quad = \text{modulus of compression} \\ &= \text{modulus of rigidity} \end{aligned} \tag{4.37}$$

This rearrangement of the terms is more than a mathematical device. These two moduli determine the velocities of the two acoustic modes, longitudinal and shear, that can propagate in an isotropic solid.

Statistical physics tells us that the Helmholtz free energy is a minimum for a system at constant temperature in thermal equilibrium. In the absence of external forces, this minimum must occur at $S_{ik} = 0$. The two quadratic forms in Equation 4.36 must be positive, so a necessary and sufficient condition for F to be positive is that $K > 0$, $\mu > 0$.

The thermodynamic relations mentioned in Section 4.4 can be used to determine the relations between stress and strain, in particular, Equation 4.32. Directly from Equation 4.36

$$\begin{aligned} \mathrm{d}F &= KS_{ll}\mathrm{d}S_{ll} + 2 \quad S_{ik} - \frac{1}{3}S_{ll}\delta_{ik} \quad \mathrm{d} \quad S_{ik} - \frac{1}{3}S_{ll}\delta_{ik} \\ &= \quad KS_{ll}\delta_{ik} + 2 \quad S_{ik} - \frac{1}{3}S_{ll}\delta_{ik} \quad \mathrm{d}S_{ik} \end{aligned} \tag{4.38}$$

so that finally

$$T_{ik} = \quad \frac{\partial F}{\partial S_{ik}} \quad _{T} = KS_{ll}\delta_{ik} + 2 \quad S_{ik} - \frac{1}{3}S_{ll}\delta_{ik} \tag{4.39}$$

Equation 4.39 shows that pure compression and shear deformation give rise to stress components proportional to K and μ, respectively. It is also a manifestation of Hooke's law, as in both cases stress is proportional to strain.

It is easy to find the inverse expression linking S_{ik} to T_{ik}. Directly from Equation 4.39

$$T_{ii} = 3KS_{ii} \tag{4.40}$$

Then immediately Equation 4.39 can be inverted to give

$$S_{ik} = \frac{\delta_{ik}T_{ll}}{9K} + \frac{\left(T_{ik} - \frac{1}{3}\delta_{ik}T_{ll}\right)}{2} \tag{4.41}$$

which again demonstrates Hooke's law. Equation 4.41 gives the important result that the diagonal components of stress and strain are uniquely connected for the case of pure hydrostatic compression. In this case, $T_{ik} = -p\delta_{ik}$ so that

$$S_{ii} = -\frac{p}{K} \tag{4.42}$$

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For small variations, we can write the compressibility χ as

$$\chi = \frac{1}{K} = -\frac{1}{V}\left.\frac{\partial V}{\partial p}\right|_T \tag{4.43}$$

Finally, Euler's theorem can be applied to obtain a compact form for F. Since F is quadratic in S_{ik}, Euler's theorem states that

$$S_{ik}\frac{\partial F}{\partial S_{ik}} = 2F \tag{4.44}$$

Together with $T_{ik} = \left(\partial F/\partial S_{ik}\right)_T$, this gives

$$F = \frac{1}{2}T_{ik}S_{ik} \tag{4.45}$$

The second approach to Hooke's law is much more direct and will be of more practical use. T_{ij} is expanded as a Taylor's series in S_{kl}:

$$T_{ij} = T_{ij}(0) + \left.\frac{\partial T_{ij}}{\partial S_{kl}}\right|_{S_{kl}=0} + \frac{1}{2}\left.\frac{\partial^2 T_{kl}}{\partial S_{ij}\partial S_{mn}}\right|_{S_{ij}=0, S_{mn}=0} S_{ij}S_{mn} \tag{4.46}$$

The first term $T_{ij}(0) = 0$ at $S_{ij} = 0$ because both stress and strain go to zero simultaneously for elastic solids. The third (nonlinear) term will be neglected here; it forms the basis of the third-order elastic constants and nonlinear acoustics. In linear elasticity, the series is truncated after the second term, leading to

$$T_{ij} = c_{ijkl}S_{kl} \tag{4.47}$$

where

$$c_{ijkl} \equiv \left.\frac{\partial T_{ij}}{\partial S_{kl}}\right|_{S_{kl}=0} \tag{4.48}$$

is known as the elastic stiffness tesnor or elastic constant tensor.

A similar Taylor's series expansion of S_{ij} in terms of T_{kl} could be carried out in identical fashion, leading to the elastic compliance tensor s_{ijkl} where

$$S_{ij} = \left.\frac{\partial S_{ij}}{\partial T_{kl}}\right|_{T_{kl}=0} T_{kl} = S_{ijkl}T_{kl} \tag{4.49}$$

Each tensor can be deduced from the other by

$$s_{ijkl} = c_{ijkl}^{-1} \tag{4.50}$$

and in what follows c_{ijkl} will be used exclusively.

Since stress is proportional to strain, c_{ijkl} represents Hooke's law in three dimensions and is the extension of the 1D spring constant k in $F = -kx$. It is obviously a fourth-rank tensor, as it must be,

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as it links two second-rank tensors. Lastly, since both T_{ij} and S_{kl} are symmetric, this symmetry is reflected in c_{ijkl}, which is also itself symmetric

$$c_{ijkl} = c_{jikl} = c_{ijlk} = c_{jilk} \tag{4.51}$$

and

$$c_{ijkl} = c_{klij} \tag{4.52}$$

These symmetry operations reduce the number of independent constants from 81 to 36 to 21 for crystals of different symmetries. The number varies from 21 (triclinic) to 3 (cubic) as is shown in numerous advanced texts in acoustics. For isotropic solids, it has already been demonstrated that there are only two independent elastic constants.

In fact it is well known [2] that for an isotropic solid

$$c_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \quad (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{4.53}$$

where λ and μ are the Lamé coefficients, which have already been introduced in Section 4.5, Equation 4.34.

It is standard practice to use a reduced notation for the elastic constants, due to the symmetry of the T_{ij} and S_{kl}. Since each of the latter has six independent components, the c_{ijkl} tensor has a maximum of 36. This leads to the introduction of the so-called engineering notation where the $c_{\alpha\beta} \equiv c_{ijkl}$. Since ij and kl go in pairs, the six α and β values are shown in Table 4.1. Again, the symmetry of c_{IJ}

$$c_{IJ} = c_{JI}$$

leads to a maximum of 21 independent constants.

Since the same symbol c is universally used for the elastic constant tensor, it is immediately obvious from the number of indices whether the full or reduced notation is being used. Thus, if c_{11} is used, it can only be in reduced notation, which is, in fact, more current in the literature. Using Hooke's law and the isotropic form of c_{ijkl}, we immediately obtain

$$T_{ij} = \lambda(S_{xx} + S_{yy} + S_{zz}) + 2 \quad {}_{ij}S_{ii} \tag{4.54}$$

for extensional stress, $i = x, y, z$ and

$$T_{ij} = 2 \quad S_{ij} \tag{4.55}$$

for tangential stress with $i, j = x, y, z$, and $i \neq j$.

TABLE 4.1
Conversion Table from Regular Indices to Reduced Indices (Engineering Notation)

α, β	*ij, kl*
1	11
2	22
3	33
4	23 = 32
5	31 = 13
6	12 = 21

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In reduced notation, the stiffness matrix in the general case is thus

$$c_{IJ} = \begin{matrix} c_{11} & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ c_{21} & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ c_{31} & c_{32} & c_{33} & c_{34} & c_{35} & c_{36} \\ c_{41} & c_{42} & c_{43} & c_{44} & c_{45} & c_{46} \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} & c_{56} \\ c_{61} & c_{62} & c_{63} & c_{64} & c_{65} & c_{66} \end{matrix} \tag{4.56}$$

whereas for the isotropic case

$$c_{IJ} = \begin{matrix} \lambda+2 & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda+2 & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda+2 & 0 & 0 & 0 \\ 0 & 0 & 0 & & 0 & 0 \\ 0 & 0 & 0 & 0 & & 0 \\ 0 & 0 & 0 & 0 & 0 & \end{matrix} \tag{4.57}$$

whereas before

$$T_J = \lambda(S_1 + S_2 + S_3) + 2\ \ S_J, \quad J = 1,2,3 \tag{4.58}$$

for extensional stress and

$$T_J = \ \ S_J, \quad J = 4,5,6 \tag{4.59}$$

for tangential stress.

4.6 OTHER ELASTIC CONSTANTS

Four other parameters have found practical use as they are directly related to measurements, which is not the case, for example, for the parameter λ for solids. Important mathematical relations between these parameters and values for representative materials are given in Tables 4.2 and 4.3.

1. Young's modulus E is defined as the ratio of axial stress to axial strain for a free-standing rod. E can be expressed using Equation 4.58 as follows.

   Let the rod be aligned along the x axis, so that the only stress component is $T_{xx} = T_1$. Then

$$\begin{aligned} T_1 &= (\lambda + 2\ \ )S_1 + \lambda(S_2 + S_3) \\ 0 &= (\lambda + 2\ \ )S_2 + \lambda(S_1 + S_3) \\ 0 &= (\lambda + 2\ \ )S_3 + \lambda(S_1 + S_3) \end{aligned} \tag{4.60}$$

   Hence

$$E = \frac{T_1}{S_1} = \frac{(3\lambda + 2\ \ )}{\lambda + } \tag{4.61}$$

   The usefulness of this parameter is that it is obtained in a standard laboratory measurement. Relations between E and the other elastic constants are given in Table 4.2; evidently, the two independent elastic constants can be chosen to be (E, μ), (E, σ), (λ, μ), or (c_{11}, c_{44}).

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TABLE 4.2
Expressions for the Elastic Constants in Terms of Different Pairs of Independent Parameters

	λ, μ	c_{11}, c_{44}	E, σ	E, μ
λ	Λ	$c_{11} - 2c_{44}$	$\dfrac{E\sigma}{(1+\sigma)(1-2\sigma)}$	$\dfrac{(E-2\ \ )}{3\ \ -E}$
μ	M	c_{44}	$\dfrac{E}{2(1+\sigma)}$	μ
E	$\dfrac{(3\lambda+2\ \ )}{\lambda+}$	$\dfrac{c_{44}(3c_{11}-4c_{44})}{c_{11}-c_{44}}$	E	E
K	$\lambda+\dfrac{2}{3}$	$c_{11}-\dfrac{4c_{44}}{3}$	$\dfrac{E}{3(1-2\sigma)}$	$\dfrac{E}{3(3\ \ -E)}$
σ	$\dfrac{\lambda}{2(\lambda+\ \ )}$	$\dfrac{c_{11}-2c_{44}}{2(c_{11}-c_{44})}$	σ	$\dfrac{E}{2}-1$

TABLE 4.3
Elastic Constants for Representative Isotropic Solids

Substance	Young's Modulus E 10^9 N·m^{-2}	Modulus of Compression K 10^9 N·m^{-2}	Lamé Constants λ 10^9 N·m^{-2}	Lamé Constants μ 10^9 N·m^{-2}	Poisson's Ratio σ
Epoxy	4.5	6.7	5.63	1.60	0.39
Lucite	3.9	6.5	5.60	1.39	0.4
Pyrex glass	60.3	39.6	23.4	24.21	0.25
PZT-5 A	104.1	94.0	67.4	39.6	0.32
Aluminum	67.6	78.1	61.4	25.0	0.36
Brass	104.8	140.2	114.7	38.1	0.38
Copper	128.6	209.0	178.2	46.0	0.40
Gold	80.6	169.1	150.1	28.4	0.42
Lead	34.7	98.8	90.8	12.1	0.44
Fused quartz	72.5	37.0	16.3	30.9	0.17
Steel	194.2	167.4	113.2	80.9	0.29
Beryllium	73.0	115.1	16.3	147.5	0.05
Sapphire (z)	895.0	298.8	201.0	145.9	0.29

2. Poisson's ratio σ is given by the ratio of the lateral contraction to the longitudinal extension of the rod in (1).

$$\sigma = -\frac{S_3}{S_1} = -\frac{S_2}{S_1} = \frac{\lambda}{2(\lambda+\ \ )} \tag{4.62}$$

σ can be measured as Young's modulus in the same experiment.

It has been pointed out by Landau and Lifshitz [2] that in principle $-1 \le \sigma \le 0.5$, although negative values of σ have never been observed. Also, it can be shown that $\sigma > 0$ corresponds to $\lambda > 0$, although neither of these is thermodynamically necessary.

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Finally, $\sigma \sim 0.5$ corresponds to materials for which the modulus of rigidity μ is small compared to the modulus of compression K.

3. Bulk modulus or modulus of compression

$$K \equiv -\frac{p}{S} \tag{4.63}$$

and its reciprocal, the compressibility

$$\chi \equiv -\frac{1}{V}\frac{\partial V}{\partial p} \tag{4.64}$$

Both parameters should be specified as being given in either adiabatic or isothermal conditions.

For a solid under uniform hydrostatic pressure

$$T_{ij} = -p\delta_{ij} \tag{4.65}$$

using

$$T_{ij} = \lambda S\delta_{ij} + 2\ S_{ij}$$

and

$$S = S_{11} + S_{22} + S_{33}$$

this gives

$$p = -\ \lambda + \frac{2}{3}\ S = -KS$$

Hence,

$$K = \lambda + \frac{2}{3} \tag{4.66}$$

as was used earlier in Equation 4.37.

4. Rigidity modulus μ. For a pure shear, $\mu \equiv$ shear stress/shear strain, for a free-standing sample. The rigidity modulus thus plays a role for shear waves analogous to that of Young's modulus for longitudinal waves in the longitudinal stretching of a free-standing rod. Since only two elastic constants are needed to describe the isotropic case fully, there are a number of possible choices. Values for each of these constants in terms of common choices for the two independent constants are given in Table 4.2. Representative values of these constants are given in Table 4.3.

SUMMARY

Tensor of order n is a tensor requiring n indices to specify it.

Einstein notation or Einstein summation convention is a convention that repeated indices in the same term of a tensor equation are summed over all available values.

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Strain tensor S_{ij} is a linearized second-order tensor describing the mechanical strain at a point. The strain tensor is symmetric.

Stress tensor *Tij* is a second-order symmetric tensor describing the local stress. The first index gives the direction of the force, the second gives the direction of the normal to the surface on which it acts.

Lamé constants λ and μ are the constants historically chosen to describe the elastic properties of an isotropic solid.

Modulus of compression or bulk modulus K is the elastic constant corresponding to hydrostatic compression.

Compressibility is the reciprocal of the bulk modulus.

Elastic constant tensor is a fourth-order symmetric tensor giving the stress tensor as a function of the strain tensor. It is also called the elastic stiffness tensor.

Young's modulus is the elastic constant corresponding to the stretching of a free-standing bar.

Poisson's ratio is the ratio of the lateral contraction to the longitudinal extension of a free-standing bar.

QUESTIONS

1. For the case of the axial extension of a bar, what would be the implications of a negative Poisson's ratio to the deformation? What would be the consequences for the other elastic parameters?
2. In Einstein notation, a spatial derivative is written using a comma, for example, $\delta u_{ij}/\delta x_j = u_{ij,j}$. Write the following differential equations and vector algebra forms in Einstein notation:
   i. grad φ
   ii. curl $\vec{\psi}$
   iii. div $\vec{E}$
   iv. $^2\vec{E}$
   v. $\dfrac{\delta^2 u}{\delta t^2} = V_0^2 \dfrac{\delta^2 u}{\delta x^2}$
3. Write out the following equations written in Einstein notation in full Cartesian form:
   i. $u_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}) + \frac{1}{2}(u_{k,i}\, u_{k,j})$
   ii. $P_i = d_{ijk}\, \delta_{jk}$
   iii. $P_i = K_0 X_{ij} E_j$
4. Verify the results of Table 4.1.
5. Write out in full the results of Equation 4.65 to show that $K = \lambda + 2\mu/3$.
   i. A rectangular plate has length l (x direction), width w (y direction), and thickness t (z direction). A uniform stress T_{xx} is applied at the ends and a uniform stress T_{yy} on both sides, so that the width remains unchanged. Using Hooke's law, determine Poisson's ratio and Young's modulus.
   ii. Express the above results as a function of E and σ.

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2. Landau, L. D. and Lifshitz, E. M., *Theory of Elasticity*, Pergamon Press, London, 1959.

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